



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CIVICS GROUP

H1 CHEMISTRY

8873/02

Paper 2 Structured Questions

16 September 2021

2 hours

Candidates answer on the Question Paper.

Additional materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use paper clips, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided

The number of marks is given in brackets [] at the end of each question or part question.

Examiner's Use		
Paper 1	MCQ	/ 30
Paper 2	Section A	
	Q1	/ 9
	Q2	/ 18
	Q3	/ 11
	Q4	/ 14
	Q5	/ 8
	Section B	
	Q6	/ 20
	Q7	/ 20
	/ 80	
Total	/ 100	
Grade		

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 Ionisation energies can be measured by finding the minimal energy required to remove the least bound atomic electrons in an evacuated tube.

- (a) Fig 1.1 shows the variation of the second ionisation energies for 9 consecutive elements R to Z with atomic numbers below 18.

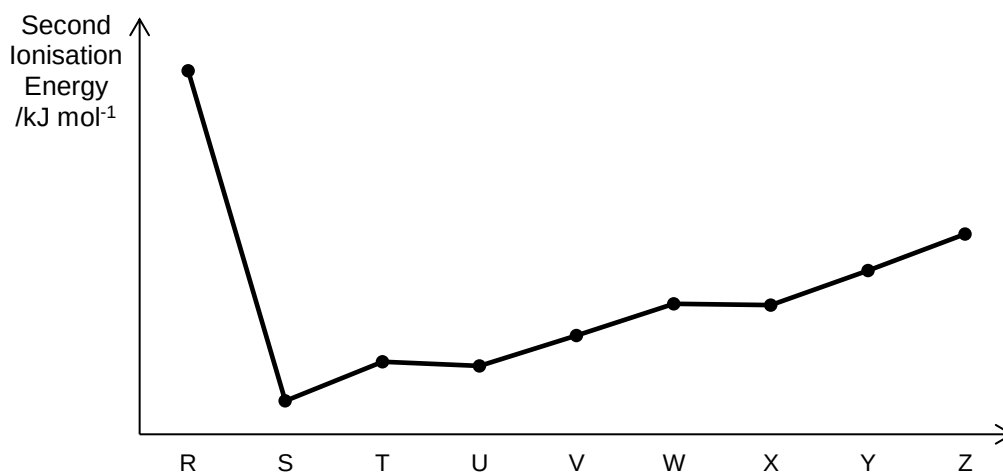


Fig. 1.1

- (i) Explain the difference in the second ionisation energy between

1. elements V and W

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2. elements W and X

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3. elements **R** and **Z**

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[5]

(ii) State the Group number for element **S**.

..... [1]

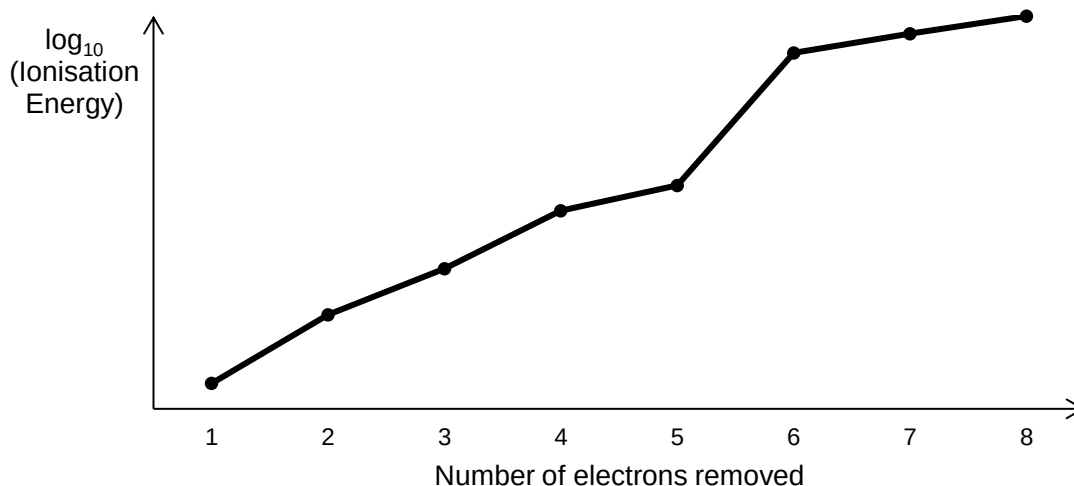
(b) Element **Q** also has atomic number below 18. Fig. 1.2 shows its first eight ionisation energies.

Fig. 1.2

(i) Deduce the Group which element **Q** belongs to.

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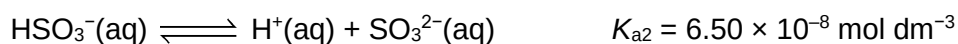
..... [2]

(ii) Hence, write the electronic configuration of element **Q**.

..... [1]

[Total: 9]

- 2 Sulfur dioxide dissolves in water to form a weak dibasic acid, H_2SO_3 , which dissociates as follows:



30.0 cm^3 of a 0.20 mol dm^{-3} solution of H_2SO_3 was titrated against a solution of 0.30 mol dm^{-3} KOH . The neutralisation process of a dibasic acid proceeds stepwise, where each equivalence point corresponds to the neutralisation of H^+ in K_{a1} and K_{a2} respectively. The resulting pH curve is shown in Fig. 2.1 below.

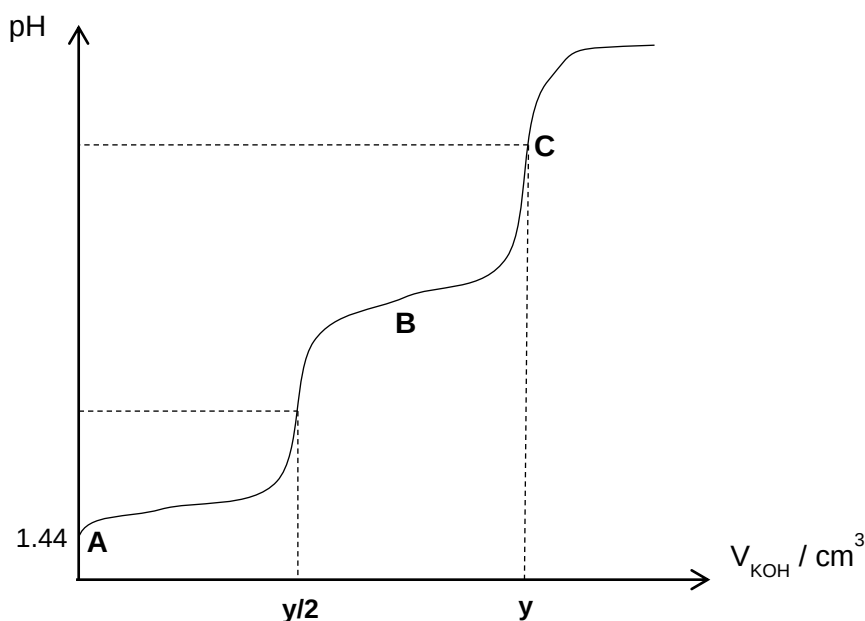


Fig. 2.1

- (a) (i) Determine the concentration of H^+ ions present at point A.

[1]

- (ii) Write the expression for the first acid dissociation constant, K_{a1} .

[1]

- (iii) Hence, calculate the value of x , the first acid dissociation constant, K_{a1} , of H_2SO_3 .

[1]

- (b) Point **B** of the titration is made up of a mixture that is able to maintain a fairly constant pH when a small amount of acid is added to it.

- (i) Identify the conjugate acid-base pair that is present at point **B**.

..... [1]

- (ii) Explain, with the aid of an equation, how this mixture at point **B** is able to maintain a fairly constant pH when a small amount of acid is added to it.

.....

 [2]

- (c) (i) Determine the equivalence volume y , at point **C**.

[2]

- (ii) Write an equation to explain why the pH at point **C** is more than 7.

..... [1]

- (iii) Suggest a suitable indicator that can be used to determine the equivalence point at point **C**.

..... [1]



- (d) In a separate experiment, a student attempted to analyse the copper content present in a sample of copper ore. First, the copper ore was dissolved in acid to convert all of the copper content into aqueous Cu^{2+} ions. $\text{HSO}_3^-(\text{aq})$ can then act as a reducing agent to reduce the $\text{Cu}^{2+}(\text{aq})$ ions present to form $\text{HSO}_4^-(\text{aq})$ and $\text{Cu}^+(\text{aq})$.

- (i) Determine the change in oxidation state of S from HSO_3^- to HSO_4^- .

.....
 [1]

- (ii) Construct the half-equation for the oxidation of HSO_3^- to HSO_4^- .

.....
 [1]

- (iii) Hence, write the overall equation for the reaction between HSO_3^- and Cu^{2+} .

..... [1]

- (iv) The $\text{Cu}^+(\text{aq})$ ions can then be determined by precipitation upon the addition of $\text{NaSCN}(\text{aq})$, leading to the formation of the highly insoluble salt $\text{CuSCN}(\text{s})$. A student obtained a 4.05 g of CuSCN precipitate from a 2.40 g of ore sample at the end of the reaction. Calculate the percentage by mass of copper in this sample of ore.

[3]



- (v) The concentration of the $\text{HSO}_3^-(\text{aq})$ solution used in the experiment was 55.7 g dm^{-3} . Using information from (iv), calculate the volume of $\text{HSO}_3^-(\text{aq})$ solution used in the experiment.

[2]

[Total: 18]



- 3 A student wanted to investigate the enthalpy change of the following reaction.



He went to the laboratory and tried looking for zinc metal and iron(II) sulfate solution to carry out the experiment. Unfortunately, the laboratory did not have any iron(II) sulfate solution. Instead, he managed to find zinc metal, iron metal and copper(II) sulfate solution. He decided to use his knowledge of Hess' Law to perform two experiments to determine ΔH_1 .

- (a) For the first experiment, he reacted excess zinc metal with $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution.



He performed the following procedure:

1. Measure 30 cm^3 of $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution into a styrofoam cup.
2. Measure the temperature for first 1.5 minutes at every 30 seconds.
3. Add all the zinc metal to the copper(II) sulfate in the styrofoam cup at exactly 2 minutes.
4. Measure the temperature at 2.5 minutes, then every 30 seconds after mixing until the temperature reaches a maximum. Thereafter, measure the temperature every 1 min after the maximum temperature has been reached.

After he recorded the timings and the temperature readings, the student plotted a graph as shown on page 9 to determine the temperature change, ΔT , using a graphical method.

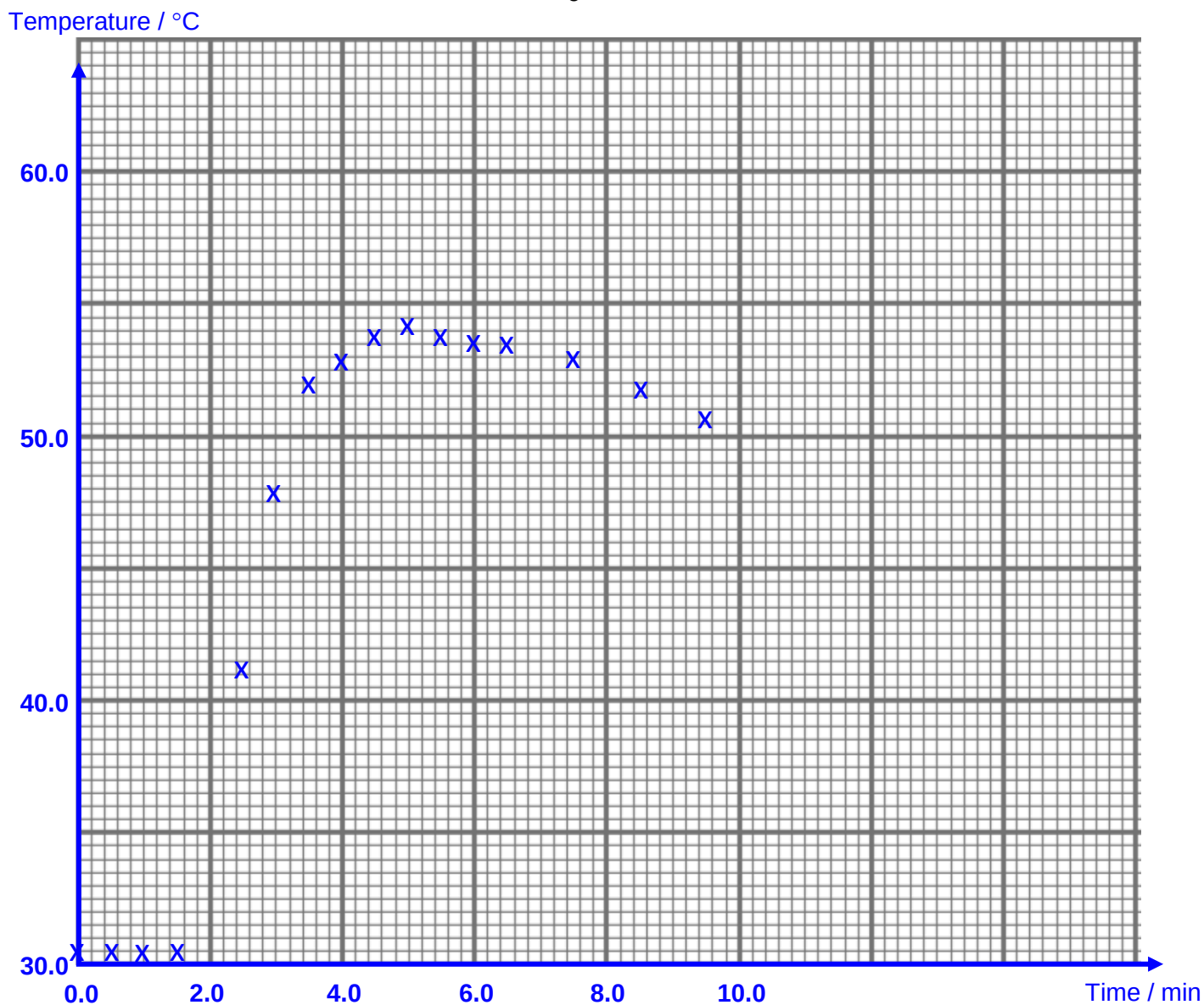


Fig 3.1

- (i) On the graph, draw two best-fit straight lines, the first taking into account the points before zinc was added, and the second taking into account the points after the reaction had finished.

Extrapolate (extend) both lines to two minutes.

Hence, determine the value of temperature change, ΔT , for ΔH_2 .

[1]

- (ii) Using your answer from (i), calculate the energy produced during the reaction of zinc with copper(II) sulfate solution.

You should assume that the specific heat capacity of the reaction mixture is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and the density of the solution is 1.00 g cm^{-3} .

[1]

- (iii) Calculate the amount of copper(II) sulfate in 30.0 cm^3 of the copper(II) sulfate solution. Hence, calculate the enthalpy change for ΔH_2 , in kJ mol^{-1} .

[3]

- (b) For the second experiment, he reacted excess iron metal with 30.0 cm^3 of $0.600 \text{ mol dm}^{-3}$ copper(II) sulfate solution and obtained the following result in the table.



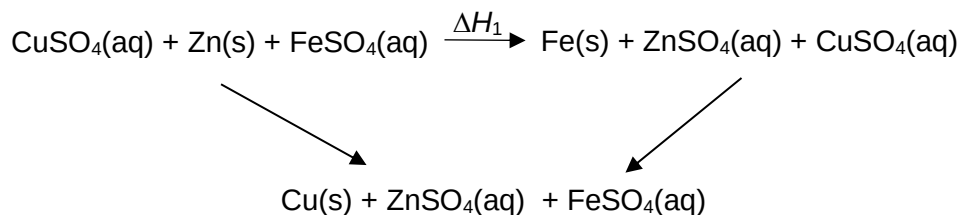
Table 3.2

Maximum temperature reached / $^{\circ}\text{C}$	35.2
Initial temperature / $^{\circ}\text{C}$	30.0

Calculate the enthalpy change for ΔH_3 , in kJ mol^{-1} .

[2]

- (c) The student constructed the energy cycle below using his knowledge of Hess' Law.



Calculate the enthalpy change for the reaction between zinc and iron (II) sulfate, ΔH_1 .

[1]

- (d) The enthalpy change of formation values for ZnSO_4 and FeSO_4 are given in Table 3.3 below.

Table 3.3

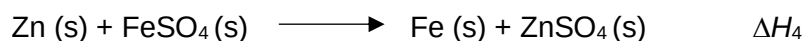
compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{ZnSO}_4(\text{s})$	-980
$\text{FeSO}_4(\text{s})$	-929

- (i) Define the term *standard enthalpy change of formation* of $\text{FeSO}_4(\text{s})$.

.....

 [1]

- (ii) Calculate ΔH_4 for the following reaction.



[2]

[Total: 11]



- 4 A reaction scheme involving butan-1-ol is shown in Fig. 4.1.

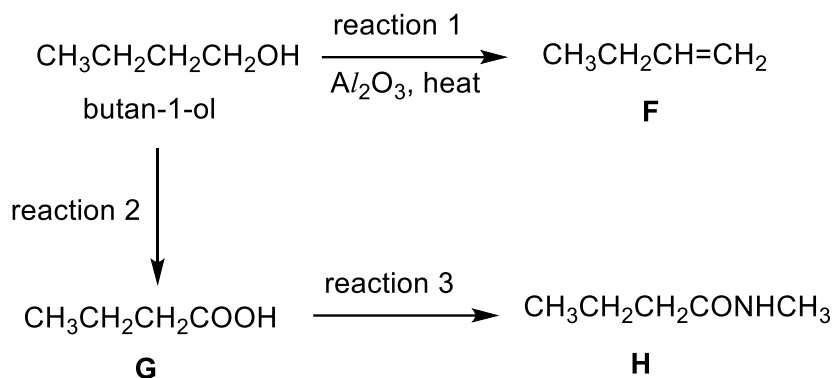


Fig. 4.1

- (a) State the type of reaction for reaction 1.
 [1]
- (b) State the reagents and conditions required for reaction 2. Write a balanced equation for the reaction that occurs. Use [O] to represent the oxidising agent.
 reagents and conditions:
 equation: [2]
- (c) State the functional group present in compound **H**, and suggest the reagents and conditions for reaction 3.
 functional group:
 reagents and conditions: [2]
- (d) Compounds **J**, **K** and **L** are isomers of compound **F**, and contain the alkene functional group. Given that compounds **J** and **K** exist as a pair of cis-trans isomers, suggest the structural formulae for compounds **J**, **K** and **L**.

J	K	L

[3]

- (e) Describe a simple chemical test which could be carried out to distinguish between compounds **G** and **H**. In your answer, state clearly the reagents and conditions used in the chemical test and give the expected observations.

.....

 [2]

- (f) (i) Write an equation to represent the standard enthalpy change of combustion for butan-1-ol, $\Delta H_c^\ominus$.

..... [1]

- (ii) With the use of relevant bond energies from the *Data Booklet*, calculate a value for the enthalpy change of combustion of butan-1-ol.

[2]

- (iii) The true value of $\Delta H_c^\ominus$ of butan-1-ol is $-2640 \text{ kJ mol}^{-1}$. Suggest a reason for the discrepancy.

.....

 [1]

[Total: 14]



- 5** Some of the most commonly used polymers are formed by the polymerisation of ethenes. The presence of side-chains affects the bulk properties of the polymer. Unbranched polymers pack closer together than polymers with several side-chains.

Poly(ethene) exists in two different forms – LDPE (low density poly(ethene)) which contains many side chains, and HDPE (high density poly(ethene)), in which there are fewer and shorter side chains.

- (a)** State the type of polymerisation that ethene undergoes in the formation of poly(ethene).

..... [1]

- (b)** Explain, with the aid of suitable diagrams, why the presence of side chains causes a difference in the densities of LDPE and HDPE.

.....
.....
.....
..... [2]

- (c)** By identifying the type of bonding between the poly(ethene) chains, explain why LDPE has a lower melting point than HDPE.

.....
..... [1]



LDPE is classified as a *thermoplastic polymer*, and is widely used in the making of grocery bags and food packaging.

For items that require more rigidity and structure such as vehicle tyres, the use of a *thermosetting polymer* is more appropriate. Thermosetting polymers can be made when polymer chains with specific side chains undergo a chemical process known as curing to produce an infusible and insoluble polymer.

- (d) With reference to their structure, bonding and properties, explain the difference between thermoplastic and thermosetting polymers.

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.....

..... [2]

- (e) Poly(ethene) bags pollute the environment for a long time because they cannot be broken down by bacteria and acids in the environment. In recent years, there is increased awareness on the need to reduce consumption of poly(ethene) by using reusable grocery bags in place of single-use ones.

- (i) Explain why poly(ethene) cannot be broken down by acids in the environment.

.....

..... [1]

- (ii) Other than reducing the use of poly(ethene), suggest another way to lessen its impact on the environment. Give your reasoning.

.....

..... [1]

[Total: 8]



Section B

Answer **one** question from this section in the spaces provided.

- 6 Naturally occurring hydrogen halides are commonly found in places with high volcanic activity. They exist as diatomic gases at room temperature and pressure.

(a) Some physical properties of three hydrogen halides are given in Table 6.1.

Table 6.1

hydrogen halide	boiling point/ °C	decomposition temperature
HCl	– 85	does not decompose
HBr	– 66	around 2000 °C
HI	– 34	350 °C

- (i) Among the hydrogen halides, HI has the highest boiling point, and yet it has the lowest decomposition temperature. By considering the bonds present, use relevant data from the *Data Booklet* to explain why this is so.

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..... [3]

- (ii) HCl and MgCl_2 both form acidic solutions when they are dissolved in water.

With the aid of equations, explain the reactions of these chlorides with water. State the expected pH of the resultant solutions formed.

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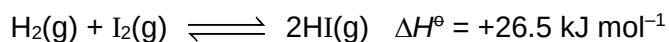
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..... [5]

Hydrogen iodide can be synthesised by reacting hydrogen with iodine. The reaction is allowed to reach an equilibrium.



- (b) The equilibrium constant, K_c , for this reaction at 200°C is 46.9.
- (i) 1 mole each of H_2 and I_2 were reacted in a 2 dm^3 vessel at 200°C . Calculate the concentration of HI that could be obtained when equilibrium was reached.

[3]



- (ii) With the use of Le Chatelier's Principle, state and explain the effect of increasing total pressure on this equilibrium.

.....
 [2]

- (iii) A sketch of the graph of $[HI]$ against time at $200\text{ }^{\circ}\text{C}$ is given in Fig. 6.2.

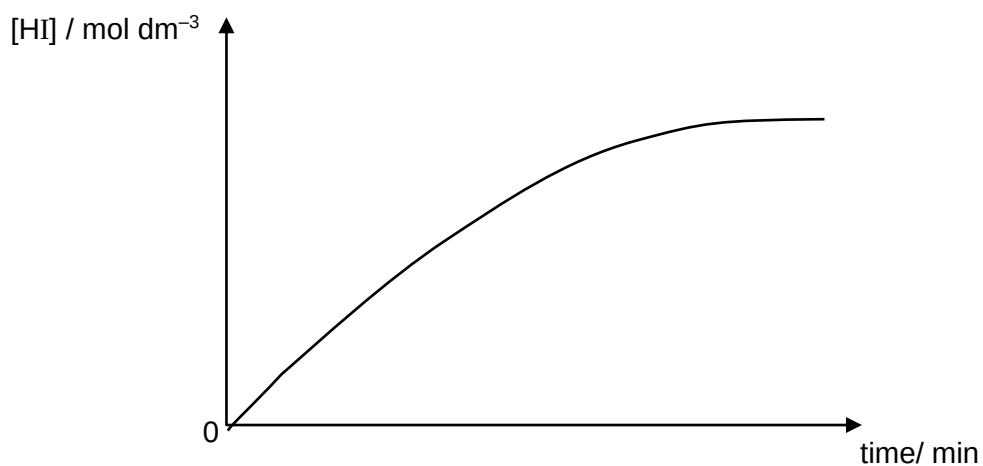


Fig. 6.2

Sketch, on the same axes, the graph if the reaction is conducted at $250\text{ }^{\circ}\text{C}$.

[2]

- (c) The rate of reaction of the synthesis of HI is studied by monitoring the concentration of I_2 with time. Two experiments, with $[H_2]$ kept constant at 0.5 mol dm^{-3} and 1.0 mol dm^{-3} are plotted as shown in Fig. 6.3.

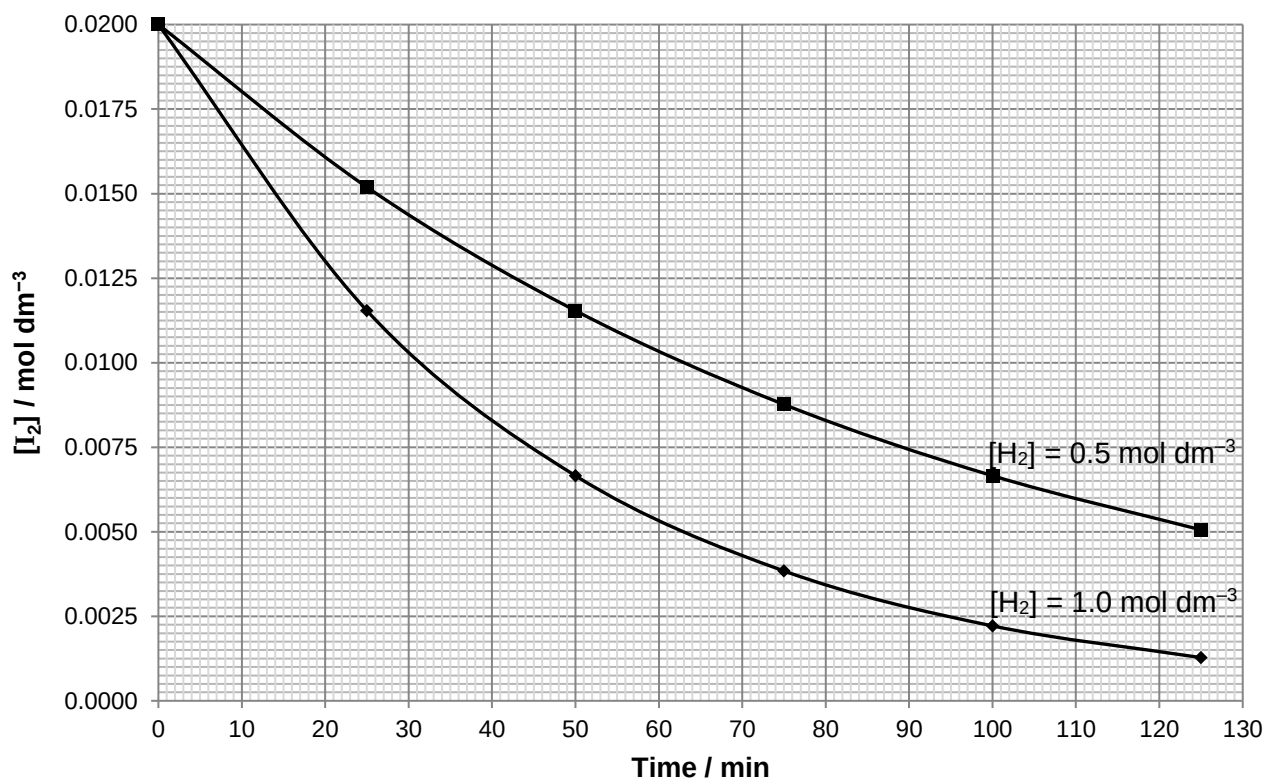


Fig. 6.3

Deduce the order of reaction with respect to I_2 and H_2 .

[4]

- (d) Hydrogen halides are commonly used in organic synthesis.

Draw the displayed formula of the major product when hydrogen iodide, HI, reacts with propene.

[1]

[Total: 20]



Reaction 1: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -197 \text{ kJ mol}^{-1}$

- | | |
|-------------|-------------------------------|
| pressure | 2 atm |
| temperature | 450 °C |
| catalyst | V ₂ O ₅ |

- the position of equilibrium,
- the rate of the reaction.

..... [5]

- (ii) At equilibrium, the concentration of SO_2 is $0.100 \text{ mol dm}^{-3}$, concentration of O_2 is $0.200 \text{ mol dm}^{-3}$ and the concentration of SO_3 is $0.0930 \text{ mol dm}^{-3}$. Write the expression for the equilibrium constant K_c and calculate the value of K_c .

[2]

- (b) The kinetics of reaction 1, the oxidation of SO_2 to produce SO_3 , is studied at constant temperature.

Table 7.1

experiment	initial concentration of SO_2 / mol dm^{-3}	initial concentration of O_2 / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.01	0.2	1.61×10^{-3}
2	0.02	0.4	6.42×10^{-3}
3	0.04	0.4	1.28×10^{-2}

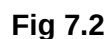
- (i) Using the data in Table 7.1, deduce the orders of reaction with respect to SO_2 and O_2 .

[2]

- (ii) Hence, write the rate equation for this reaction.

..... [1]





(c) Although SO_2 is an oxide of a Period 3 element, its chemical and physical properties differ from some of the other oxides, like magnesium oxide, MgO , and aluminium oxide, Al_2O_3 .

- Shape:

Bond angle:

(ii) Write equations for any reactions that these three oxides would have with hydrochloric acid and with sodium hydroxide. Identify any of these oxides that shows no reaction.

..... [5]



- (iii) Explain, in terms of structure and bonding, why SO_2 has a much lower melting point than MgO .

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..... [2]

[Total: 20]

